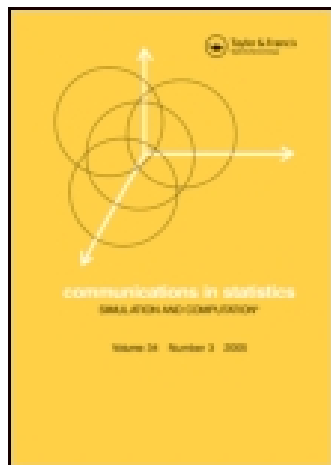


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An Empirical Analysis of Some Nonparametric Goodness-of-Fit Tests for Censored Data

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In this article, we consider some nonparametric goodness-of-fit tests for right censored samples, viz., the modified Kolmogorov, Cramer–von Mises–Smirnov, Anderson–Darling, and Nikulin–Rao–Robson χ^2 tests. We also consider an approach based on a transformation of the original censored sample to a complete one and the subsequent application of classical goodness-of-fit tests to the pseudo-complete sample. We then compare these tests in terms of power in the case of Type II censored data along with the power of the Neyman–Pearson test, and draw some conclusions. Finally, we present an illustrative example.

Keywords Anderson–Darling test; Censored samples; Cramer–von Mises–Smirnov test; Empirical power; Goodness-of-fit; Kolmogorov test; Monte Carlo simulations; Neyman–Pearson test; Nikulin–Rao–Robson chi-squared test; Pseudo-complete sample.

Mathematics Subject Classification 62G10.

1. Introduction

In reliability and survival studies, the observed data are usually of the form

$$X = (X_1, \delta_1), (X_2, \delta_2), \dots, (X_n, \delta_n),$$

where $X_i = \min(T_i, C_i)$ is the observation value, T_i is the lifetime, and C_i is a censoring time. The censoring indicator $\delta_i = 1$ if $X_i = T_i$ and $\delta_i = 0$ if $X_i = C_i$.

There are various forms of right censoring (for details, see Balakrishnan and Cohen, 1991; Cohen, 1991):

- If individuals are observed up to some pre-fixed time C , then censoring is referred to as Type I censoring and in this case when $\delta_i = 0 : X_i = C$.

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- If a life test is terminated whenever a specified number of failures (say, r) has occurred, it is referred to as Type II censoring and in this case when $\delta_i = 0$: $X_i = X_{(r)}$, where $X_{(r)}$ is the last observed failure time.
- Let the lifetime T and the censoring time C be independent random variables from distribution functions $F(t)$ and $F^C(t)$, respectively. All lifetimes and censoring times are assumed to be mutually independent. Then, this form of censoring is referred to as independent random censoring.

In this article, we consider the problem of testing the simple goodness-of-fit hypothesis $H_0: F = F_0$ and the composite hypotheses, which can be presented as $H_0: F \in \{F_0(\cdot; \theta), \theta \in \theta\}$.

The modification of the classical Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests for Type I and Type II censored data are given in Barr and Davidson (1973), D’Agostino and Stephens (1986), Pettitt and Stephens (1976), and Reineke and Crown (2004). In the case of randomly censored data, these tests can be modified with the use of the Kaplan–Meier estimate instead of the empirical distribution function; see, for example, Hjort (1992), Koziol and Green (1976), Nair (1981), and Reineke and Crown (2004). In Chimitova and Tsivinskaya (2011), Chimitova et al. (2011), and Lemeshko et al. (2010a), the distributions of the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling statistics have been studied by means of Monte Carlo simulations. In Lemeshko et al. (2010a), the authors have proposed approximations for the limiting distributions of the test statistics in the case of Type I and Type II censored samples, and have also carried out a power study.

Another group of goodness-of-fit tests for censored data contains the χ^2 -type tests; see, for example, Akritas (1988), Bagdonavicius et al. (2010), Habib and Thomas (1986), Hjort (1992), Hollander and Pena (1992), and Kim (1993). The idea of comparing observed and expected numbers of failures in time intervals was originally discussed in Barr and Davidson (1973) and Hjort (1992). In Bagdonavicius et al. (2010), this idea has been developed by considering the choice of random grouping intervals as data functions and then presenting simple formulas for computing the test statistics for many common lifetime distributions. The chi-squared goodness-of-fit test was also developed for parametric accelerated failure time models; see, for example, Bagdonavicius et al. (2013).

In this article, we discuss some advantages and disadvantages of the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests, as well as the tests based on pseudo-complete samples and the Nikulin–Rao–Robson (NRR) χ^2 test. We compare these tests by means of power when testing simple and composite hypotheses, and we evaluate them against the Neyman–Pearson test. The comparative study is through Monte Carlo simulations. From the obtained results, we draw some conclusions. Finally, we present an example to illustrate the discussed methods.

2. Nonparametric Goodness-of-Fit Tests for Censored Data

2.1. Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling Tests

One approach for testing goodness-of-fit hypotheses based on complete samples is through nonparametric tests such as Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests. The Kolmogorov test statistic is given by

$$D_n = \sup_{-\infty < t < \infty} |F_n(t) - F_0(t; \theta)|,$$

where $F_n(t)$ is the empirical distribution function. In practice, the statistic is usually used with the Bolshev correction (Bolshev and Smirnov, 1983) of the form

$$S_K = \frac{6n \cdot D_n + 1}{6\sqrt{n}}. \quad (1)$$

The Cramer–von Mises–Smirnov test statistic is given by

$$S_\omega = n \int_{-\infty}^{\infty} (\hat{F}_n(t) - F_0(t; \theta))^2 dF_0(t; \theta). \quad (2)$$

The Anderson–Darling test statistic can be presented as

$$S_\Omega = n \int_{-\infty}^{\infty} (\hat{F}_n(t) - F_0(t; \theta))^2 \frac{dF_0(t; \theta)}{F_0(t; \theta)(1 - F_0(t; \theta))}. \quad (3)$$

Let us denote the distribution of a test statistic under H_0 by $G(S|H_0)$. In the case of testing a simple hypothesis, the distributions $G(S|H_0)$ of the considered statistics do not depend on the tested distribution. Statistic S_K belongs to the Kolmogorov distribution, while S_ω and S_Ω belong to $a1$ and $a2$ distributions, respectively; see Bolshev and Smirnov (1983). For composite hypotheses, the distributions of the nonparametric test statistics $G(S|H_0)$ are affected by a number of factors: the form of the tested lifetime distribution $F_0(t; \theta)$, the type and the number of estimated parameters, the method of parameter estimation used, and so on. Approximations for the limiting distributions of the test statistics for testing various composite hypotheses have been discussed in Lemeshko and Lemeshko (2009a, 2009b) and Lemeshko et al. (2010b).

The Kaplan–Meier estimate of the lifetime distribution is used instead of the empirical distribution $F_n(t)$ in the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests based on censored samples. In this case, the test statistics in (1), (2), and (3) are calculated on the interval $[0, \tau]$, where $\tau = \max_{1 \leq i \leq n} \{\delta_i X_i\}$ is the last observed failure time.

In Chimitova and Tsivinskaya (2011), Chimitova et al. (2011), and Lemeshko et al. (2010a), the distributions of the modified statistics have been studied for Type I, Type II, and randomly censored samples. The distributions of the test statistics based on the Kaplan–Meier estimate depend critically on the form and degree of censoring. In the case of randomly censored samples, the distributions of these test statistics also depend on the distribution of censoring times, which is usually unknown, and for this reason, we propose to use the Kaplan–Meier estimate of the distribution of censoring times for simulation of a randomly censored sample.

Thus, in order to simulate the same censoring scheme as in the original sample, one can use the following algorithm:

1. Generate a complete sample of size n according to the distribution F_0 : $T_i = F_0^{-1}(\zeta_i; \theta)$, where ζ_i is uniformly distributed as $U(0, 1)$, $i = 1, 2, \dots, n$.
2. Invert the original censored sample by changing the values δ_i to $1 - \delta_i$.
3. Calculate the Kaplan–Meier estimate of the distribution of the censoring time $\hat{F}^C(t)$ by the inverted sample obtained in Step 2.

4. Generate uniform $U(0, 1)$ variates and calculate the values $C_i, i = 1, 2, \dots, n$. Then,
 - a. if $\xi_i < \hat{F}^C(c_1)$, set $C_i = \frac{\xi_i \cdot c_1}{\hat{F}^C(c_1)}$;
 - b. if $\xi_i \in (\hat{F}^C(c_j), \hat{F}^C(c_{j+1}))$, set $C_i = c_j + \frac{(\xi_i - \hat{F}^C(c_j)) \cdot (c_{j+1} - c_j)}{(\hat{F}^C(c_{j+1}) - \hat{F}^C(c_j))}, j = 1, 2, \dots, r$;
 - c. if $\xi_i > \hat{F}^C(c_r)$, set $C_i = c_r + c_r(\xi_i - \hat{F}^C(c_r))$, where $c_1, \dots, c_r$ are the increasingly ordered different censoring times in the original sample and r is the number of different censoring times in the original sample.
5. Finally, set $X_i = \min(T_i, C_i)$ and $\delta_i = 1 \{T_i \leq C_i\}, i = 1, 2, \dots, n$.

So, for testing goodness of fit with the modified tests and calculating the corresponding p -values, it is necessary to simulate the distribution of the test statistics under the null hypothesis. However, the process of simulation requires time and computational cost and it is not possible without special software. This is a significant disadvantage of the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests for censored samples. Moreover, in the case of random censoring, the distributions of modified test statistics depend on the degree of censoring, which in turn is defined by the combination of distributions $F(t)$ and $F^C(t)$. If the distribution $F_0(t)$ corresponding to the tested hypothesis H_0 is close enough to the unknown true distribution $F(t)$, then the average degree of censoring among the simulated samples will be close to the degree of censoring in the original sample, otherwise the usage of the modified tests will face some problems.

2.2. Nikulin–Rao–Robson χ^2 Test

Chi-squared-type tests require dividing an observed interval $[0, \tau]$ into k smaller intervals $I_j = (a_{j-1}, a_j), a_0 = 0, a_k = \tau$. Denote by $U_j = \sum_{X_i \in I_j} \delta_i$ the number of observed failures and by e_j the “expected” number of failures in the interval $I_j, j = 1, \dots, k$. Then, the NRR χ^2 test statistic (Bagdonavicius et al., 2010) can be written in the form

$$Y^2 = Z^T \hat{V}^- Z, \quad (4)$$

where $Z = (Z_1, \dots, Z_k)^T, Z_j = \frac{1}{\sqrt{n}}(U_j - e_j), \hat{V}^-$ is the generalized inverse of the matrix $\hat{V}, \hat{V}^- = \hat{A}^{-1} + \hat{A}^{-1} \hat{C}^T \hat{G}^- \hat{C} \hat{A}^{-1}, \hat{A}$ is the $k \times k$ diagonal matrix with diagonal elements $\hat{A}_j = \frac{U_j}{n}, j = 1, \dots, k, \hat{G} = [\hat{g}_{ll'}]_{m \times m}, \hat{g}_{ll'} = \hat{t}_{ll'} - \sum_{j=1}^k \hat{C}_{lj} \hat{C}_{l'j} \hat{A}_j^{-1}, \hat{t}_{ll'} = \frac{1}{n} \sum_{\delta_i=1}^n \frac{\partial \ln \lambda(X_i; \hat{\theta})}{\partial \theta_l} \frac{\partial \ln \lambda(X_i; \hat{\theta})}{\partial \theta_{l'}}, \hat{C}_{lj} = \frac{1}{n} \sum_{\delta_i=1}^n \frac{\partial \ln \lambda(X_i; \hat{\theta})}{\partial \theta_l},$ with $\lambda(t)$ being the hazard rate function and $\hat{\theta}$ being the maximum likelihood estimates of the unknown parameters.

In Bagdonavicius et al. (2010), it is recommended to take a_j as random data functions in order to divide the interval $[0, \tau]$ into k intervals with equal expected numbers of failures. So, a_j can be calculated as follows: define $E_k = \sum_{i=1}^n \Lambda(X_i; \hat{\theta}), b_i = (n - i) \Lambda(X_{(i)}; \hat{\theta}) + \sum_{l=1}^i \Lambda(X_{(l)}; \hat{\theta}), E_j = \frac{j}{k} E_k, j = 1, \dots, k$, with $X_{(0)} = 0$. If i is the smallest natural number such that $E_j \in [b_{i-1}, b_i], j = 1, \dots, k - 1$, then

$$\hat{a}_j = \Lambda^{-1}((E_j - \sum_{l=1}^i \Lambda(X_{(l)}; \hat{\theta})) / (n - i + 1), \hat{\theta}), \hat{a}_k = X_{(n)},$$

where Λ^{-1} is the inverse of the cumulative hazard function Λ .

The limiting distribution of the test statistic is χ_q^2 , where $q = \text{rank}(\hat{V}^-)$. Thus, the null hypothesis is rejected with approximate significance level α if $Y^2 > \chi_\alpha^2(r)$.

If a simple hypothesis is tested, the test statistic has the following form:

$$Y^2 = \sum_{j=1}^k \frac{(U_j - e_j)^2}{U_j}.$$

The choice of $\hat{a}_j$ is the same as in the composite hypothesis case, and we just need to skip $\hat{\theta}$ in all the above formulas. The limiting distribution of the test statistic is χ_k^2 . So, the hypothesis is rejected with approximate significance level α if $Y^2 > \chi_\alpha^2(k)$. For more details on this goodness-of-fit test and its properties and performance, one may refer to the recent book (Voinov et al., 2013).

The NRR χ^2 test has a number of advantages as compared with the modified nonparametric tests described earlier. In particular, there is no significant dependence of the distribution of the NRR statistic on the distribution of censoring times. Also, as the sample size grows, the distribution of the NRR statistic converges to the corresponding χ^2 distribution. Moreover, it is possible to choose the number of grouping intervals and boundary points for considered pair of competing hypotheses in order to increase the power of the test. Nevertheless, it is necessary to mention here that the calculation of this statistic is rather complicated.

3. Application of Nonparametric Goodness-of-Fit Tests Based on Pseudo-Complete Samples

In this section, we discuss the transformation of the original censored sample to a complete sample and then to apply the classical Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests for the so obtained pseudo-complete sample.

3.1. Simple Hypothesis

Let us consider a simple hypothesis, viz., when F_0 is completely specified. For each censored observation $(X_i, \delta_i = 0)$, a value $\hat{T}_i$ is generated as $\hat{T}_i = F_0^{-1}(\xi_i)$, where $\xi_i \sim U[F_0(C_i), 1)$. The corresponding values of censored observations in the original sample are replaced by these generated values. Hereby, we obtain a pseudo-complete sample $\hat{X}_1, \hat{X}_2, \dots, \hat{X}_n$, in which

$$\hat{X}_i = \begin{cases} X_i, & \text{if } \delta_i = 1 \\ \hat{T}_i, & \text{if } \delta_i = 0. \end{cases}$$

Under the simple null hypothesis, the distribution of the completed sample is F_0 . So, the algorithm for testing the simple hypothesis is as follows:

1. Specify the significance level α .
2. Transform the original censored sample X to a pseudo-complete sample $\hat{X}_1, \hat{X}_2, \dots, \hat{X}_n$.
3. Calculate the test statistic S^* ((1), (2), or (3)) from this pseudo-complete sample and the specified distribution F_0 .
4. Compute the p -value as $p = 1 - G(S^*|H_0)$.
5. The hypothesis H_0 is rejected if the obtained p -value is less than the pre-specified α .

Through Monte Carlo simulations, we have shown that the distributions of the test statistics based on the pseudo-complete samples, when testing simple hypothesis, do not depend on the distribution $F^C(t)$ and coincide with the corresponding limiting distributions (e.g., the Kolmogorov distribution for Kolmogorov's test statistic) (Chimitova et al., 2011).

3.2. Composite Hypothesis

Let $\hat{\theta}$ be the maximum likelihood estimate of the unknown parameter θ determined from the original censored sample. We then replace all censored observations X_i with $\delta_i = 0$ by simulated times $\hat{T}_i = F_0^{-1}(\xi_i)$, where ξ_i is uniformly distributed in the interval $[F(C_i, \hat{\theta}), 1)$. We thus obtain a pseudo-complete sample $\hat{X}_1, \hat{X}_2, \dots, \hat{X}_n$.

After replacing the censored observations by generated observations with values $\hat{T}_i$, it is necessary to estimate θ again. The algorithm of testing a composite hypothesis based on this pseudo-complete sample proceeds as follows:

1. Specify the significance level α .
2. Calculate the maximum likelihood estimate $\hat{\theta}$ from the original censored sample.
3. Transform the original censored sample to a complete sample $\hat{X}_1, \hat{X}_2, \dots, \hat{X}_n$.
4. Calculate the maximum likelihood estimate $\tilde{\theta}$ from this pseudo-complete sample.
5. Calculate the test statistic S^* ((1), (2), or (3)) from this sample and the distribution $F_0(\cdot; \tilde{\theta})$.
6. Compute the p -value as $p = 1 - G(S^* | H_0)$.
7. The hypothesis H_0 is rejected if the obtained p -value is less than the pre-specified α .

By Monte Carlo simulations, we have once again shown that the distributions of the test statistics based on pseudo-complete samples, when testing composite goodness-of-fit hypotheses, do not differ from the corresponding approximations of the limiting distributions, if the degree of censoring is not high (no more than about 30%) (Chimitova et al., 2011).

So, in the case of testing a simple goodness-of-fit hypothesis with censored samples including randomly censored samples, it is possible to use the classical Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests, and to calculate the corresponding p -values based on the Kolmogorov, $a1$ and $a2$ limiting distributions, respectively. However, when testing a composite hypothesis, it is necessary to take into account the number of censored observations, since if the degree of censoring is large, then the use of the approximations obtained from complete samples to calculate the p -value will be inaccurate.

4. Empirical Power Study

In Lemeshko et al. (2010a), the power of the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests were all investigated based on Type II censored samples, and they were compared with the NRR χ^2 test in Chimitova and Tsivinskaya (2011) in terms of power. The power of classical Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests for samples completed from original randomly censored samples has been studied in Chimitova et al. (2011). There, it has been shown that the power of the considered tests increases with the sample size. It is important to note that in the case of Type I and random censoring, we cannot simulate the same censoring scheme for both hypotheses H_0 and H_1 , and it may not be appropriate to compare different tests by power in such situations.

Table 1

Power of the tests for the lognormal-generalized Weibull pair when testing the simple hypothesis

Test	Degree of censoring								
	0%	5%	10%	20%	30%	40%	50%	60%	70%
S_K^M	0.29	0.28	0.29	0.26	0.24	0.21	0.17	0.14	0.11
S_ω^M	0.32	0.31	0.30	0.25	0.22	0.17	0.14	0.11	0.10
S_Ω^M	0.35	0.32	0.30	0.25	0.22	0.19	0.17	0.15	0.14
S_K^R	0.29	0.28	0.29	0.27	0.26	0.23	0.19	0.15	0.12
S_ω^R	0.32	0.32	0.32	0.29	0.27	0.24	0.20	0.16	0.13
S_Ω^R	0.35	0.34	0.33	0.30	0.27	0.23	0.20	0.17	0.14
$Y^2_{,3}$	0.29	0.24	0.21	0.18	0.15	0.13	0.11	0.10	0.10
$Y^2_{,5}$	0.18	0.15	0.14	0.11	0.10	0.10	0.10	0.10	0.10

In this article, we compare the considered goodness-of-fit tests by power for four different pairs of competing hypotheses through Monte Carlo simulations. The estimates of the power are calculated from the distributions of the test statistics $G(S|H_0)$ and $G(S|H_1)$, which are simulated based on Type II censored samples of size $n = 200$. In the case of composite hypotheses, the parameters are estimated by the maximum likelihood method. The number of simulations used is $N = 10^5$. The values of the power of the tests are calculated with the nominal significance level $\alpha = 0.1$.

4.1. Lognormal-Generalized Weibull Pair of Competing Hypotheses

The null hypothesis H_0 is the lognormal distribution with distribution function $F(x; \theta_1, \theta_2) = \frac{1}{2} - \frac{1}{2\sqrt{\pi}} \Gamma\left(-\frac{1}{2\theta_2^2} \log^2\left(\frac{x}{\theta_1}\right), \frac{1}{2}\right)$ and the parameters $\theta_1 = 2$, $\theta_2 = 2$. The competing hypothesis H_1 is the generalized Weibull distribution with distribution function $F(x; \theta_1, \theta_2, \theta_3) = 1 - \exp(1 - (1 + (\frac{x}{\theta_1})^{\theta_2})^{-\theta_3})$ and parameters $\theta_1 = 0.35$, $\theta_2 = 1$, $\theta_3 = 3.36$. The distribution functions of these models do not intersect, but both hazard rate functions decrease.

In Table 1, the powers of the considered tests are presented for various degrees of censoring in the case of testing the simple hypothesis. The modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests are denoted by S_K^M , S_ω^M , and S_Ω^M , respectively, while the classical Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests based on pseudo-complete samples are denoted by S_K^R , S_ω^R , and S_Ω^R , respectively, and the NRR χ^2 test is denoted by Y^2 with the number of intervals k .

As can be seen from Table 1, the power estimates of all tests decrease as the degree of censoring increases. The classical tests based on pseudo-complete samples have higher power than the corresponding modified tests, and the Anderson–Darling is the most powerful among the considered tests. For this pair of competing hypotheses, the power of the NRR χ^2 test is observed to be the lowest and for large degrees of censoring it is even equal to the significance level.

In Table 2, the powers of the considered tests are given in the case of testing the composite hypothesis.

Table 2

Power of the tests for the lognormal-generalized Weibull pair when testing the composite hypothesis

Test	Degree of censoring								
	0%	5%	10%	20%	30%	40%	50%	60%	70%
S_K^M	0.15	0.15	0.15	0.15	0.16	0.16	0.16	0.17	0.17
S_ω^M	0.16	0.16	0.16	0.16	0.17	0.17	0.18	0.18	0.18
S_Ω^M	0.19	0.19	0.19	0.19	0.20	0.20	0.21	0.21	0.21
S_K^R	0.15	0.15	0.15	0.15	0.15	0.14	0.14	0.12	0.11
S_ω^R	0.16	0.17	0.17	0.16	0.17	0.16	0.15	0.13	0.12
S_Ω^R	0.19	0.19	0.19	0.19	0.19	0.18	0.17	0.15	0.13
$Y^{2,3}$	0.17	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15
$Y^{2,5}$	0.15	0.14	0.14	0.13	0.14	0.13	0.13	0.13	0.13

As can be seen from Table 2, the modified Anderson–Darling test turns out to be the most powerful one in comparison with the Cramer–von Mises–Smirnov, Kolmogorov, and NRR tests. For censoring degrees up to 20%, the power of the modified and classical Anderson–Darling tests are almost equal, but when the degree of censoring is more than 20%, the power of classical tests based on pseudo-complete samples decreases, whereas the power of the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests all increase. The power of the NRR χ^2 test practically does not change with the censoring degree growth. It is important to note that in the case of $k = 5$, the power of the NRR χ^2 test is less than in the case of $k = 3$.

4.2. Exponential-Weibull Pair of Competing Hypotheses

The null hypothesis H_0 is the exponential distribution with distribution function $F(x; \theta_1) = 1 - \exp(-\frac{x}{\theta_1})$ and parameter $\theta_1 = 1$. The competing hypothesis H_1 is the Weibull distribution with distribution function $F(x; \theta_1, \theta_2) = 1 - \exp(-(\frac{x}{\theta_1})^{\theta_2})$ and parameters $\theta_1 = 1, \theta_2 = 1.2$. The distribution functions of these models intersect once, and the exponential distribution has constant hazard rate and the hazard rate function of the Weibull distribution increases.

In Table 3, the powers of the considered tests are presented for various degrees of censoring in the case of testing the simple hypothesis.

In contrast to the previous pair of competing hypotheses in the case of testing simple hypothesis, the power is not monotone decreasing. The power of tests decreases, but after some level of degree of censoring it increases. Moreover, for large degrees of censoring, the power of the modified tests turns out to be even larger than the power of these tests in the case of complete samples. The most powerful test among the considered ones is the NRR χ^2 test, but for large degrees of censoring (more than 50%), the modified Anderson–Darling and Cramer–von Mises–Smirnov tests become better than the NRR χ^2 test.

In Table 4, the powers of the considered tests are presented in the case of testing the composite hypothesis.

When the composite hypothesis is tested, the power of all considered tests decrease with increasing number of censored observations. The modified Anderson–Darling test has higher power than other nonparametric tests. It is interesting to note that, between the

Table 3
Power of the tests for the exponential-Weibull pair when testing the simple hypothesis

Test	Degree of censoring								
	0%	5%	10%	20%	30%	40%	50%	60%	70%
S_K^M	0.38	0.39	0.37	0.36	0.35	0.37	0.42	0.49	0.58
S_ω^M	0.42	0.40	0.37	0.35	0.37	0.43	0.50	0.59	0.67
S_Ω^M	0.56	0.49	0.44	0.42	0.45	0.51	0.56	0.62	0.68
S_K^R	0.38	0.39	0.38	0.36	0.35	0.34	0.34	0.35	0.37
S_ω^R	0.42	0.43	0.40	0.36	0.33	0.33	0.35	0.38	0.40
S_Ω^R	0.56	0.55	0.50	0.43	0.39	0.39	0.41	0.43	0.45
$Y^2_{,3}$	0.76	0.70	0.65	0.58	0.53	0.52	0.52	0.53	0.57
$Y^2_{,5}$	0.72	0.66	0.61	0.55	0.52	0.50	0.49	0.51	0.53

classical tests based on pseudo-complete samples, the Anderson–Darling test is the best once again. The power estimates of the NRR χ^2 test are seen to be higher for the lower number of intervals ($k = 3$).

4.3. Weibull-Generalized Weibull Pairs of Competing Hypotheses

In this case, we consider two pairs of competing hypotheses. The null hypothesis H_0 is the Weibull distribution with parameters $\theta_1 = 2, \theta_2 = 2$. In the case of the first pair, the competing hypothesis H_1 is the generalized Weibull distribution with parameters $\theta_1 = 2, \theta_2 = 2, \theta_3 = 0.9$. In the case of the second pair, the competing hypothesis H_1 is the generalized Weibull distribution with parameters $\theta_1 = 2, \theta_2 = 2, \theta_3 = 1.1$.

In Tables 5 and 6, the powers of the considered tests are presented for various degrees of censoring in the case of testing the simple hypothesis.

As can be seen from Table 5, for the first pair of competing hypotheses, when the generalized parameter θ_3 is equal to 0.9, the most powerful test is the classical Anderson–Darling test based on pseudo-complete samples. The power of the NRR χ^2 test is very low in this

Table 4
Power of the tests for the exponential-Weibull pair when testing the composite hypothesis

Test	Degree of censoring								
	0%	5%	10%	20%	30%	40%	50%	60%	70%
S_K^M	0.84	0.80	0.76	0.69	0.62	0.55	0.47	0.40	0.33
S_ω^M	0.90	0.87	0.83	0.76	0.69	0.61	0.53	0.45	0.38
S_Ω^M	0.91	0.88	0.85	0.79	0.72	0.65	0.56	0.47	0.38
S_K^R	0.84	0.79	0.73	0.63	0.54	0.45	0.35	0.26	0.16
S_ω^R	0.90	0.86	0.81	0.71	0.61	0.50	0.38	0.27	0.17
S_Ω^R	0.91	0.88	0.84	0.74	0.65	0.55	0.45	0.35	0.24
$Y^2_{,3}$	0.83	0.78	0.74	0.66	0.59	0.51	0.44	0.37	0.30
$Y^2_{,5}$	0.79	0.74	0.69	0.62	0.54	0.47	0.39	0.33	0.27

Table 5

Power of the tests for the first Weibull-generalized Weibull pair when testing the simple hypothesis

Test	Degree of censoring								
	0%	5%	10%	20%	30%	40%	50%	60%	70%
S_K^M	0.49	0.49	0.49	0.45	0.40	0.34	0.27	0.22	0.16
S_ω^M	0.55	0.53	0.50	0.43	0.35	0.29	0.22	0.16	0.11
S_Ω^M	0.58	0.53	0.49	0.41	0.33	0.28	0.22	0.18	0.14
S_K^R	0.49	0.49	0.49	0.47	0.43	0.37	0.30	0.23	0.16
S_ω^R	0.55	0.54	0.54	0.50	0.44	0.38	0.31	0.24	0.17
S_Ω^R	0.58	0.57	0.55	0.49	0.43	0.36	0.29	0.23	0.18
$Y^2_{,3}$	0.51	0.45	0.41	0.32	0.26	0.20	0.15	0.12	0.08
$Y^2_{,5}$	0.35	0.29	0.26	0.19	0.15	0.11	0.09	0.07	0.05

case. Comparing the modified nonparametric tests with each other, we note that as the degree of censoring becomes larger than 20%, the Kolmogorov test has higher power than the modified Anderson–Darling and Cramer–von Mises–Smirnov tests, which was not the case with the other pairs of competing hypotheses.

As seen from Table 6, when the generalized parameter is equal to 1.1 and the degree of censoring is less than 30%, the most powerful test is the NRR χ^2 test with number of intervals $k = 3$. For the degrees of censoring larger than 30%, this test becomes less powerful than other nonparametric tests. In the case of small degrees of censoring, the modified Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests have almost the same power as the corresponding classical tests based on pseudo-complete samples. However, for large degrees of censoring, the modified nonparametric tests become more powerful than other tests.

Table 6

Power of the tests for the second Weibull-generalized Weibull pair when testing the simple hypothesis

Test	Degree of censoring								
	0%	5%	10%	20%	30%	40%	50%	60%	70%
S_K^M	0.43	0.43	0.43	0.43	0.41	0.38	0.34	0.30	0.26
S_ω^M	0.48	0.48	0.48	0.46	0.42	0.39	0.35	0.31	0.28
S_Ω^M	0.56	0.55	0.53	0.47	0.41	0.36	0.31	0.27	0.23
S_K^R	0.43	0.43	0.43	0.42	0.40	0.37	0.32	0.26	0.20
S_ω^R	0.48	0.47	0.47	0.46	0.42	0.38	0.32	0.26	0.20
S_Ω^R	0.56	0.55	0.53	0.48	0.43	0.37	0.31	0.25	0.20
$Y^2_{,3}$	0.64	0.59	0.55	0.48	0.42	0.37	0.31	0.27	0.23
$Y^2_{,5}$	0.60	0.55	0.52	0.44	0.39	0.34	0.29	0.26	0.21

The estimates of the power of all the considered tests in the case of the composite hypotheses turn out to be equal to the significance level ($\alpha = 0.1$), i.e., none of the considered tests can distinguish between the Weibull distribution and the generalized Weibull distribution in this case, which can be explained as follows: for the generalized Weibull distribution with the chosen parameter values, it is possible to fit Weibull distribution that quite closely approximates it.

5. Illustrative Example

Consider the initial remission times of leukemia patients (in days) induced by three types of treatment (with $n_1 = 25$, $n_2 = 19$, and $n_3 = 22$). In Lee and Wang (2003), it has been shown that the data do not show significant differences among the three initial treatments (the hypothesis of homogeneity is not rejected at significance level $\alpha = 0.05$). This means that the observations can be combined into one independent random sample (with $n = 66$) as follows:

(4, 1), (5, 1), (9, 1), (10, 1), (12, 1), (13, 1), (10, 1), (8, 1), (10, 1), (10, 1),
 (12, 1), (14, 1), (8, 1), (10, 1), (11, 1), (23, 1), (25, 1), (25, 1), (23, 1),
 (28, 1), (28, 1), (28, 1), (29, 1), (20, 1), (48, 1), (70, 1), (75, 1), (99, 1),
 (103, 1), (28, 1), (28, 1), (31, 1), (31, 1), (40, 1), (31, 1), (32, 1), (37, 1),
 (41, 1), (41, 1), (162, 1), (169, 1), (195, 1), (220, 1), (48, 1), (89, 1),
 (124, 1), (143, 1), (57, 1), (62, 1), (74, 1), (100, 1), (139, 1), (161, 0),
 (199, 0), (217, 0), (12, 0), (159, 0), (190, 0), (196, 0), (20, 0), (258, 0),
 (269, 0), (245, 0), (197, 0), (205, 0), (219, 0)

Now, let us consider testing the goodness of fit of the inverse Gaussian distribution by the nonparametric tests discussed in the preceding sections. The CDF of the inverse Gaussian distribution is

$$F_{IG}(t; \theta_1, \theta_2, \theta_3) = \frac{1}{2} + \frac{\exp(2\theta_2/\theta_3)}{2} + \Gamma\left(\frac{\theta_1\theta_2}{2t(t/(\theta_1\theta_3) - 1)^2}; \frac{1}{2}\right) / (2\sqrt{\pi}) \\ + \exp(2\theta_2/\theta_3)\Gamma\left(\frac{\theta_1\theta_2}{2t(t/(\theta_1\theta_3) + 1)^2}; \frac{1}{2}\right) / (2\sqrt{\pi}),$$

where $\Gamma(a; b)$ is the incomplete gamma-function, $t > 0$, $\theta_1 > 0$, $\theta_2 > 0$, $\theta_3 > 0$.

Let the significance level α be set as 0.05. The maximum likelihood estimates of the unknown parameters are $\hat{\theta}_1 = 64.58$, $\hat{\theta}_2 = 0.45$, and $\hat{\theta}_3 = 2.93$. In this case, we computed the values of the considered test statistics based on the sample of the initial remission times. In Table 7, the test statistics and the corresponding p -values are presented.

In the case of the modified nonparametric tests, the p -values were estimated based on the simulated distributions $G(S|H_0)$ using the algorithm described earlier in Section 2.1. To calculate the p -values of the classical Kolmogorov, Cramer-von Mises-Smirnov, and Anderson-Darling tests, we used the approximations of the limiting distributions for the considered hypothesis proposed in Lemesko et al. (2010b). The p -values of the NRR test were obtained by the chi-squared distribution with k degrees of freedom.

Table 7

Test statistics when testing goodness-of-fit of the inverse Gaussian distribution for remission times and the corresponding p -values

Test	Statistic	p -Value
Modified Kolmogorov	0.56	0.72
Modified Cramer–von Mises–Smirnov	0.05	0.53
Modified Anderson–Darling	0.27	0.55
Classical Kolmogorov	0.61	0.69
Classical Cramer–von Mises–Smirnov	0.05	0.70
Classical Anderson–Darling	0.32	0.70
Nikulin–Rao–Robson, $k = 3$	1.60	0.67
Nikulin–Rao–Robson, $k = 5$	7.56	0.26

The obtained p -values are larger than the specified significance level, and therefore there is no evidence to reject the null hypothesis H_0 for these data.

6. Discussion

While carrying out a comparative analysis of different goodness-of-fit tests, it is quite natural to compare the considered tests with the most powerful test. According to the Neyman–Pearson lemma, in the case of distinguishing between two models, one of which (the null model) is a special case of the other (the alternative model) and each of which has no unknown parameters, the likelihood ratio test has the highest power among all competitors. The likelihood ratio statistic is given by

$$\Lambda(X) = \frac{L(X|H_1)}{L(X|H_0)}, \quad (5)$$

where $L(X) = \prod_{i=1}^n f^{\delta_i}(X_i)(1 - F(X_i))^{1-\delta_i}$ is the likelihood function for censored sample $X = (X_1, \delta_1), (X_2, \delta_2), \dots, (X_n, \delta_n)$. Usually, it is more convenient to consider the logarithm of the statistic (5). Let α be the specified significance level; then, H_0 is rejected if $\Lambda(X) > B_\alpha$, where B_α is such that $P(\Lambda(X) > B_\alpha | H_0) = \alpha$.

In some particular cases, the distribution of the Neyman–Pearson test statistic can be found analytically, but in most cases we need to use Monte Carlo simulations.

So, in the case of the simple hypothesis $H_0 : F(x) = F_1(x; \theta = \theta_0)$ against the simple alternative $H_1 : F(x) = F_1(x; \theta = \theta_1)$, the algorithm of simulating the distribution $G(\Lambda(X) | H_0)$ is as follows:

1. Generate a sample of observations according to the null hypothesis
2. Calculate the likelihood functions $L(X|H_0)$ and $L(X|H_1)$ and the value of the test statistic in (5).

By repeating the above process N times, a random sample from the distribution of the test statistic is generated.

Table 8

Power of the Neyman–Pearson test for the exponential-Weibull pair and relative percentage of power of the best nonparametric test

Censoring degrees		0%	5%	10%	20%	30%	40%	50%	60%	70%
Simple	Λ	0.99	0.97	0.96	0.92	0.88	0.84	0.82	0.80	0.80
hypotheses	Relative percentage	77%	72%	68%	63%	61%	61%	69%	77%	84%
Composite	Λ	0.99	0.97	0.96	0.92	0.88	0.84	0.82	0.80	0.80
hypotheses	Relative percentage	77%	72%	68%	63%	61%	61%	69%	77%	84%

Now, let us consider the composite hypothesis $H_0 : F(x) = F_1(x; \theta \in \Omega_0)$ against the composite alternative $H_1 : F(x) = F_1(x; \theta \in \Omega_1)$, where $\Omega_0 \subset \Omega_1$, $\Omega_1 \setminus \Omega_0 \neq \emptyset$. Then, the algorithm of simulating the distribution $G(\Lambda(X)|H_0)$ can be given as follows:

1. Generate a sample of observations according to the null hypothesis.
2. Estimate the unknown parameters by maximizing the likelihood function $L(X|H_0)$ with respect to $\theta \in \Omega_0$.
3. Estimate the unknown parameters by maximizing the likelihood function $L(X|H_1)$ with respect to $\theta \in \Omega_1$.
4. Calculate the statistic in (5) using $L(X|H_0)$ and $L(X|H_1)$ obtained in Steps 2 and 3, respectively.

By repeating the above process N times, a random sample from the distribution of the test statistic is generated.

Since any parametric goodness-of-fit test including the Neyman–Pearson test have the information of the competing hypothesis, the power of such tests is known to be higher than the power of nonparametric goodness-of-fit tests. However, for complete investigation of nonparametric tests, it would be interesting to evaluate the loss incurred in power due to the absence of information of competing hypothesis. In other words, let us estimate the ratio of the power of the best nonparametric test among the considered tests to the power of the Neyman–Pearson test.

In Table 8, the estimates of the power of the Neyman–Pearson test are presented in the case of the exponential-Weibull pair of competing hypotheses. The relative percentage of power of the best nonparametric test to the power of the Neyman–Pearson test is also given in the table.

As seen from Table 8, the losses of the power of nonparametric tests are between 16% and 29% when testing the simple hypothesis and 5%–18% in the case of composite hypothesis. Knowledge of the kind of competing hypothesis H_1 does not give a significant advantage, especially in the case of composite hypotheses, which is indeed the typical situation in practice. This fact strongly supports the use of nonparametric tests in this context.

7. Concluding Remarks

So, with regard to the question “Which goodness-of-fit test is most preferable for right censored samples?,” there is no single test that would be the best in any situation. Nevertheless, the results of our investigation allow us to state some advantages and disadvantages of the considered tests. As to the modified Kolmogorov, Cramer–von Mises–Smirnov, and

Anderson–Darling tests, in the case of Type I and Type II censored data, one can use the tables of percentage points for these tests in simple hypothesis testing (Lemeshko et al. 2010a)). While testing a composite hypothesis, it is possible to simulate the distribution of test statistics for the distribution under test and the given censoring scheme. The modified Anderson–Darling test seems to be the most powerful in most cases.

It is also found to be useful to consider the transformation of a censored sample to a complete sample and then to apply the classical Kolmogorov, Cramer–von Mises–Smirnov, and Anderson–Darling tests based on the pseudo-complete sample. In the case of small degrees of censoring, when there is no significant bias in the parameter estimates, it is possible to use the approximations of the limiting distributions obtained in Kim (1993) for the calculation of p -value while testing composite hypotheses. These tests tend to have at least as much power as the corresponding modified tests.

The NRR test has many advantages compared to the considered nonparametric tests. In particular, there is no significant dependence of the distribution of the NRR statistic (in the case of limited sample sizes) on the distribution of censoring times. As the sample size grows, the distribution of the NRR statistic converges to the corresponding χ^2 distribution. Here, we considered this test as a nonparametric test, but for some given pair of competing hypotheses, it is possible to find the optimal number of grouping intervals and boundary points that maximize the power of the test.

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